
Air Pollution Prediction from Environmental Data

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Abstract

This study predicts the reference carbon monoxide concentration, $\text{CO}(\text{GT})$, from the UCI *Air Quality* dataset. The sample contains $n \simeq 9,000$ hourly records from one urban monitoring station: pollutant concentrations, sensor responses, and meteorological variables. We replaced the code -200 with missing values, removed $\text{NMHC}(\text{GT})$ because it was mostly missing, and filled the remaining gaps by linear interpolation over time. After removing rows with an unknown target, about 7,600 usable observations remained. We then split the data into training and test sets. Every model-dependent step, especially standardization, was fitted on training data only and refitted inside each cross-validation fold. Linear interpolation was done before the split, so a small risk of temporal leakage remains. The complete Python workflow is available in the project repository [1]. We compared Ridge regression, Lasso regression, and k -nearest neighbors. Five-fold cross-validation selected the hyperparameters on the training set. The separate test set gave RMSE, MAE, and R^2 . The strongest links with $Y = \text{CO}(\text{GT})$ came from $\text{C6H6}(\text{GT})$, $\text{PT08.S2}(\text{NMHC})$, and $\text{PT08.S1}(\text{CO})$. The meteorological variables had much smaller linear correlations. KNN gave the best test scores, with $R^2 \simeq 0.937$ and $\text{RMSE} \simeq 0.362$. Its fit is very local: $k = 5$, Manhattan distance, and distance weights. This makes the result sensitive to noise, measurement error, and nearby hourly records. A random split can also place similar hours in both sets. Ridge and Lasso remain useful because their coefficients are more stable and easier to read.

Keywords: air pollution; carbon monoxide; statistical learning; Ridge regression; Lasso regression; KNN; cross-validation; air quality.

1 Introduction

Air pollution is a major public health issue. Long-term exposure is linked to respiratory and cardiovascular disease and to a higher risk of early death [3,4]. Modern sensors now record pollution and weather at fine time scales. Hourly measurements make it possible to study daily cycles, short pollution peaks, and links between several pollutants. These data are well suited to statistical learning.

We use the *Air Quality* dataset from the UCI Machine Learning Repository [2]. The target is the reference carbon monoxide concentration, $Y = \text{CO}(\text{GT})$. Carbon monoxide is linked to combustion and road traffic, so it is a useful marker of urban air pollution. The prediction problem is $\hat{Y} = f(X)$, where X contains the other measurements. This study estimates predictive relations; it does not estimate causal effects or physical mechanisms. The study has two goals: (i) build a complete workflow

for environmental data, from cleaning and imputation to exploratory analysis, train-test splitting, standardization, and validation; and (ii) compare Ridge, Lasso, and KNN. Ridge and Lasso are regularized linear baselines that can handle correlated predictors. KNN has no fixed linear form and tests whether local nonlinear structure improves prediction [7,8]. We compare predictive accuracy, stability, sensitivity to the validation design, and interpretability.

2 Dataset and Exploratory Analysis

2.1 Dataset structure

The dataset has $n \simeq 9,000$ hourly observations, with one row for each recorded hour, and three variable groups: (i) reference, or *Ground Truth*, concentrations $\text{CO}(\text{GT})$, $\text{C6H6}(\text{GT})$, $\text{NOx}(\text{GT})$, and $\text{NO2}(\text{GT})$; (ii) sensor responses $\text{PT08.S1}(\text{CO})$, $\text{PT08.S2}(\text{NMHC})$, $\text{PT08.S3}(\text{NOx})$, $\text{PT08.S4}(\text{NO2})$, and $\text{PT08.S5}(\text{O3})$; and (iii) weather variables: temperature T , relative humidity RH , and absolute humidity AH . The first two groups measure pollution directly or through sensor signals, while the third describes atmospheric conditions.

The target is $Y = \text{CO}(\text{GT})$, a combustion-related pollutant that is important in urban areas. All other variables form the predictor vector X .

2.2 Data quality overview

Missing values are stored as -200 , which is outside the physical range of these measurements and cannot be used as a real numeric value. In addition, $\text{NMHC}(\text{GT})$ is missing at most times, and the target $\text{CO}(\text{GT})$ is missing for some rows. A supervised model needs a known target for every training observation, so these entries must be handled before model fitting.

An interquartile-range rule flagged several extreme values. We kept them when there was no clear recording error. Many may be real pollution peaks, and they matter when the goal is to predict high-pollution hours. Automatic deletion would remove rare observations that may be important for the final task.

2.3 Temporal dynamics of pollution

The hourly profiles have a clear 24-hour cycle. $\text{CO}(\text{GT})$, $\text{NOx}(\text{GT})$, $\text{NO2}(\text{GT})$, and $\text{C6H6}(\text{GT})$ have two main peaks: one near 8 a.m. and one between 6 p.m. and 8 p.m. Concentrations fall between these periods. This timing matches morning and evening traffic, although no direct traffic variable is present in the dataset.

Weather follows another cycle: temperature usually peaks in the afternoon, and relative humidity falls as temperature rises. Absolute humidity changes more slowly over the day. These weather cycles do not match the two

Hourly distribution of pollutants and meteorological variables

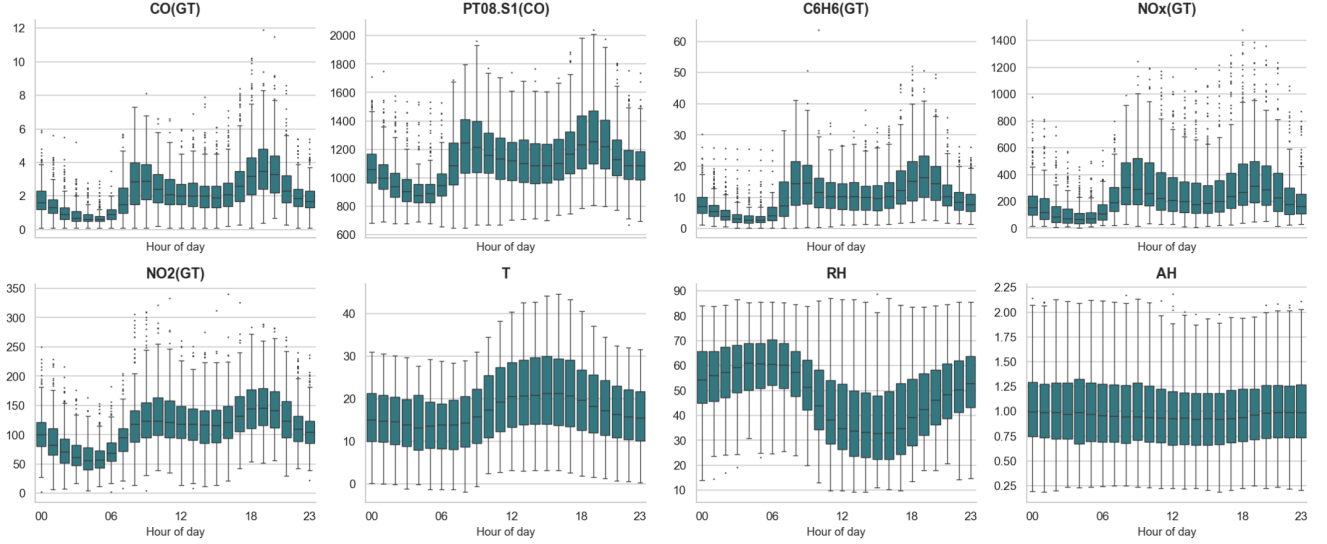


Figure 1: Hourly pollutant and weather cycles. Traffic-related variables have two peaks, near 8 a.m. and 6–8 p.m. The timing matches rush hours; the figure shows a temporal pattern only.

pollution peaks, so their simple hourly profiles provide less direct information about traffic-related emissions.

Pollution levels are also higher on weekdays than at weekends. Higher traffic, commuting, and urban activity from Monday to Friday can explain this pattern. The dataset contains no direct intervention or traffic count, so it shows a temporal association only.

2.4 Correlations with the target variable

Let $Y = \text{CO}(\text{GT})$. The largest positive correlations are $\rho(Y, \text{C6H6}(\text{GT})) \simeq 0.910$, $\rho(Y, \text{PT08.S2}(\text{NMHC})) \simeq 0.895$, and $\rho(Y, \text{PT08.S1}(\text{CO})) \simeq 0.863$. The strongest negative value is $\rho(Y, \text{PT08.S3}(\text{NOx})) \simeq -0.696$. For T , RH , and AH , the marginal linear correlations are only about 0.005–0.058. Each value measures the linear link between one predictor and the target after cleaning. It does not include interactions or time delays. Weather can still act through nonlinear, delayed, or conditional effects.

Several predictors are also strongly correlated with each other. This is expected: combustion pollutants share sources, and related sensors respond to the same mixtures. In ordinary least squares, collinearity can make small data changes produce large coefficient changes. Ridge and Lasso add penalties that reduce this sensitivity, so they are natural choices here [5,6].

3 Data Preprocessing

3.1 Cleaning and imputation

We mapped every value -200 to missing, removed $\text{NMHC}(\text{GT})$ because it was mostly missing, and deleted rows with an unknown target $\text{CO}(\text{GT})$. Rows without Y cannot contribute to a supervised loss function. The sample size changed from $n \simeq 9,000$ to $n \simeq 7,600$, while the remaining predictor columns were kept for modeling.

We filled the remaining gaps by linear interpolation along

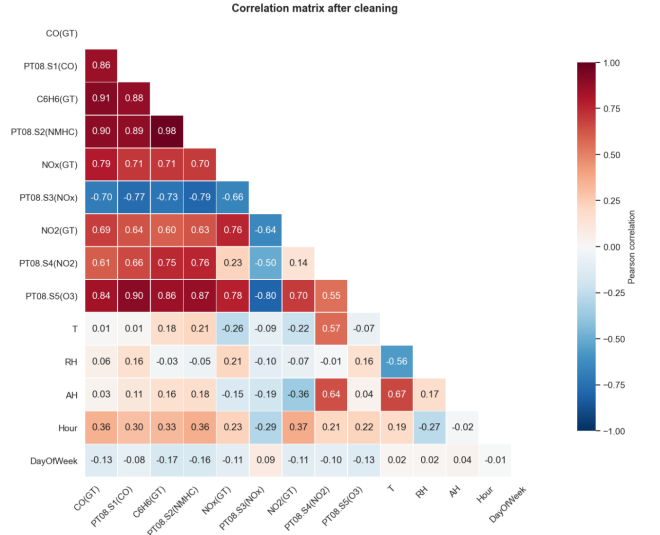


Figure 2: Correlation matrix after cleaning. Pollutant and sensor variables have large correlations with $\text{CO}(\text{GT})$; weather variables have small marginal linear correlations.

time. For two known times $t_0 < t < t_1$, a missing value at t is estimated by $x_t = x_{t_0} + \frac{t-t_0}{t_1-t_0}(x_{t_1} - x_{t_0})$. Hourly pollution and weather often change smoothly between adjacent records, so this gives simple local estimates. It also uses nearby temporal information and can flatten sharp pollution spikes.

3.2 Train-test split and validation protocol

The cleaned data were split into 80% for training and 20% for testing. Training data were used for fitting and hyperparameter selection; the test set was kept for the final comparison. In 5-fold cross-validation, 4/5 of the training data fitted the model and 1/5 evaluated it. This

was repeated so that every fold served once as validation data. For each candidate hyperparameter set, the five validation scores were averaged, and the best mean score selected the final configuration. The test set remained untouched until the last evaluation [8].

All model-dependent preprocessing stayed inside the validation workflow. In each fold, the scaler was fitted on the training part and applied to the validation part. For the final test, it was fitted on the full training set and applied once to the test set. The validation and test observations never affected the scaling parameters. This prevents leakage from scaling. Linear interpolation was done before the split, so a small amount of temporal information may cross between the two sets. Interpolation is the only main preprocessing step outside the model pipeline. A stricter design would impute inside each fold or use chronological validation.

The observations are time-indexed, but the main benchmark uses a random split after cleaning and interpolation. Adjacent hours can then appear in both sets, and their pollution and sensor values may be very similar. This is especially important for KNN because it predicts from nearby points. The random design can favor this local model and make the scores slightly optimistic. A blocked or chronological scheme would train on earlier observations and test on later ones, which better matches future prediction. The reported values are random-split benchmark scores.

3.3 Standardization

Each predictor X_j was standardized inside the validation pipeline after the train-test split:

$$X_{\text{std}} = \frac{X - \mu_{\text{train}}}{\sigma_{\text{train}}},$$

where μ_{train} and σ_{train} come only from the current training data.

Scaling matters for all three methods. In Ridge and Lasso, the penalty acts on the coefficient scale, so comparable feature scales make the same value of α meaningful across variables. In KNN, distances combine all coordinates. A variable with a large numerical range would dominate the distance if the data were not standardized.

4 Modeling Methods

We compare Ridge, Lasso, and KNN. Ridge estimates a global linear relation and shrinks all coefficients. Lasso also uses a global linear relation and can set some coefficients to zero. KNN estimates a local nonlinear relation directly from neighboring observations and makes no global parametric assumption. The comparison covers predictive accuracy, coefficient stability, sensitivity to local noise, and interpretability.

4.1 Ridge regression

Ridge solves the following ℓ_2 -penalized least-squares problem [5]:

$$\min_{\beta} \left\{ \sum_{i=1}^n (y_i - x_i^\top \beta)^2 + \alpha \sum_{j=1}^p \beta_j^2 \right\}.$$

The parameter α controls shrinkage: $\alpha = 0$ gives ordinary least squares, while $\alpha \uparrow$ usually gives $\|\hat{\beta}\|_2 \downarrow$. Coefficient variance falls, which helps under collinearity, while bias can rise when the penalty becomes too strong.

Cross-validation selected $\alpha \simeq 1.87$ [1]. On the test set, Ridge gave RMSE $\simeq 0.465$ and $R^2 \simeq 0.896$. Thus the linear model explained about 89.6% of the test-set variation around the mean. It captured most of the signal while keeping one coefficient for each predictor.

4.2 Lasso regression

Lasso uses an ℓ_1 penalty. This penalty can set some coefficients exactly to zero and select variables [6]:

$$\min_{\beta} \left\{ \sum_{i=1}^n (y_i - x_i^\top \beta)^2 + \alpha \sum_{j=1}^p |\beta_j| \right\}.$$

Cross-validation selected $\alpha = 0.001$ [1]. This weak penalty gave $\beta_j \neq 0$ for every predictor, so all variables remained in the fitted model. Test scores were almost the same as Ridge: RMSE $\simeq 0.466$, MAE $\simeq 0.295$, and $R^2 \simeq 0.896$. Lasso selected no variables here. With highly correlated predictors, several variables carry similar information, and the model can move weight from one predictor to another across samples. Sparse solutions can therefore change even when prediction stays stable. The coefficients measure associations, not causal effects.

4.3 k -nearest neighbors

KNN predicts a response from nearby points in feature space and does not impose a global linear equation [7]. With uniform weights, the estimate is the mean of the k nearest responses:

$$\hat{y}(x) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(x)} y_i,$$

where $\mathcal{N}_k(x)$ is the set of the k closest observations. With distance weights, the prediction becomes

$$\hat{y}(x) = \frac{\sum_{i \in \mathcal{N}_k(x)} w_i(x) y_i}{\sum_{i \in \mathcal{N}_k(x)} w_i(x)},$$

with $w_i(x) \propto d(x, x_i)^{-1}$. The closest points then contribute most.

Five-fold cross-validation tested several KNN configurations [1]. The best one used $k = 5$, Manhattan distance, which corresponds to $p = 1$, and distance weights. On the test set, it gave RMSE $\simeq 0.362$, MAE $\simeq 0.228$, and $R^2 \simeq 0.937$, better than both linear models on all three metrics.

The gain suggests useful local nonlinear structure. Each prediction uses only five neighbors, so noise, measurement error, and temporal proximity can have a large

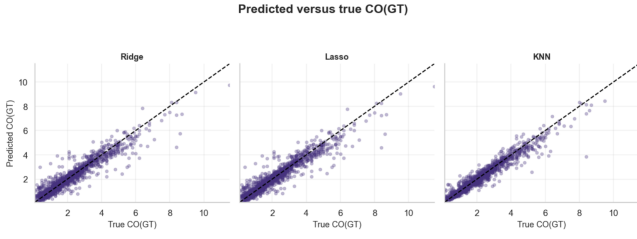


Figure 3: True and predicted CO(GT) values. KNN lies closest to the ideal diagonal. Its very local fit can still overfit.

effect. Distance weights increase the influence of the closest records. A random split can place adjacent hourly observations in both sets and strengthen this advantage. KNN is the best model for this protocol, but the result is not a general ranking for every station or period. Repeated splits, blocked validation, and external data are needed to test its robustness.

5 Model Evaluation

For residuals $e_i = y_i - \hat{y}_i$, the evaluation used RMSE, MAE, and R^2 :

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}.$$

Table 1: Test-set scores. Hyperparameters were selected by cross-validation on training data only. For KNN, $p = 1$ means Manhattan distance and distance means distance-based weighting.

Model	RMSE	MAE	R^2
Ridge	0.465	0.296	0.896
Lasso	0.466	0.295	0.896
KNN ($k = 5$, $p = 1$, distance)	0.362	0.228	0.937

Table 1 shows that KNN has the lowest RMSE and MAE and the highest R^2 . Its RMSE is about 22% lower than the Ridge value. Ridge and Lasso are nearly identical on every metric, so both capture the same main linear signal and the Lasso penalty adds little predictive change.

These scores come from one held-out split at one monitoring station. They do not guarantee the same accuracy at another station, in another city, or during another period. With $k = 5$, each KNN prediction depends on a very small region of the training sample. KNN is therefore more sensitive than Ridge and Lasso to noise, temporal dependence, distribution shifts, and changes in sensor behavior.

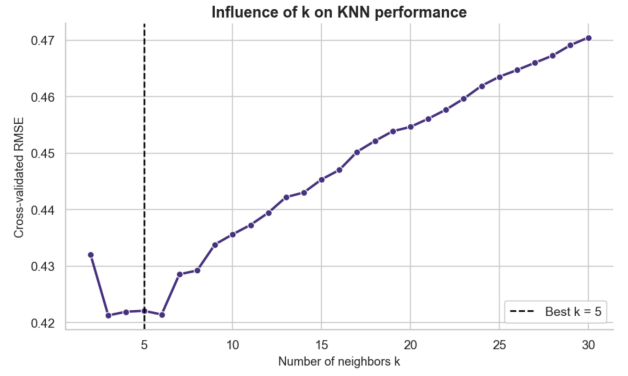


Figure 4: KNN error as a function of k . Small neighborhoods give the lowest error here, and the error rises when k becomes large. The local choice $k = 5$ needs external validation.

5.1 Influence of hyperparameters

For Ridge and Lasso, α controls bias and variance. When $\alpha \downarrow 0$, correlated coefficients can remain unstable. When α is too large, the coefficients are overshrunk and the model underfits. Cross-validation searches for a value between these two cases.

For KNN, k sets the neighborhood size; the metric and weights set each neighbor’s influence. In general, $k \downarrow$ gives lower bias and higher variance, while $k \uparrow$ smooths the prediction, lowers variance, and raises bias. Manhattan distance uses $d(x, z) = \sum_j |x_j - z_j|$, and distance weights emphasize the closest points. The selected configuration ($k = 5$, Manhattan distance, and distance weights) uses fine local structure but may have high variance.

5.2 Stability of the results

Ridge and Lasso are stable because regularization controls coefficient variance under strong predictor correlations. Both solve convex optimization problems, so the fitted solution is reproducible for fixed data and hyperparameters. Their nearly equal scores show that they use the same main linear signal.

KNN depends more on the geometry of the feature space, the distance, the weighting rule, and k . Standardization gives comparable scales, but $k = 5$ uses only five records for each prediction, and distance weights can make one or two records dominate. Consecutive hours may be almost identical. The test set shows useful local structure, but a single split gives limited evidence. Repeated random splits, blocked validation, or another dataset are needed for a stronger stability claim.

6 Interpretation and Discussion

6.1 Most important variables

The strongest correlations with CO(GT) come from C6H6(GT), PT08.S2(NMHC), PT08.S1(CO), and PT08.S3(NO_x). These variables track combustion emissions and sensor responses at the same station, so they provide direct information about the target.

Weather variables have weak marginal correlations, but they can still affect pollutant dispersion, accumulation, and dilution through nonlinear, delayed, or conditional paths. A small Pearson value only describes the simplest linear relation.

6.2 Model interpretability

Ridge and Lasso are the easiest models to explain. After standardization, each coefficient gives the sign and relative size of the association between a predictor and $C0(GT)$. Ridge fits this dataset well because it stabilizes correlated coefficients, keeps all predictors, and changes smoothly when the data change.

Lasso can select a variable by setting $\hat{\beta}_j = 0$. Here $\alpha = 0.001$ is very small and every coefficient remains nonzero. All predictors are retained, so the validation results do not support a smaller or clearly sparser model.

KNN gives the highest predictive accuracy and the least direct explanation. It has no global coefficients or single equation linking each variable to the target. Each prediction is determined by nearby observations, so interpretation requires inspection of the selected neighbors and their distances.

6.3 Statistical interpretation versus causality

The models estimate $E[Y | X]$, not the effect of changing X . A large association between $C6H6(GT)$ and $C0(GT)$ does not mean that benzene produces carbon monoxide. Both pollutants can come from traffic and combustion, which creates a common-source association. A small weather correlation means only a weak marginal linear association; it does not remove a possible physical role.

A causal study would need a precise intervention, explicit control of confounders, and more data on traffic, industrial activity, wind, season, and emission sources. It would also need a validation design linked to that causal question. The current random split and predictive loss cannot answer it. This project is a prediction and model-comparison study.

6.4 Comparison between prediction and understanding

For test-set prediction, KNN is the strongest model. It can use local nonlinear patterns among sensors, pollutants, and the target after standardization. For scientific interpretation, Ridge and Lasso are clearer because their coefficients refer directly to the measured variables and summarize one global relation. Ridge gives the best balance between interpretability and stability under multicollinearity. All observations come from one station, so the results may depend on local traffic, urban structure, sensor position, season, and study period. Tests on other stations or later periods are needed before broad use.

6.5 Limitations

The main limits are: (i) all data come from one urban station, so local conditions can shape every result; (ii) performance uses one random test split, while repeated or blocked temporal splits would give a stronger stability

check; (iii) $C0(GT)$ and several predictors are pollutant measurements from the same station, which makes the task easier than prediction from low-cost sensors and weather alone; and (iv) interpolation was done before the split and may leak a small amount of temporal information. These points mainly affect how far the test scores can be generalized.

7 Conclusion

Environmental measurements and sensor responses provide a strong signal for $Y = C0(GT)$. The workflow uses cleaning, interpolation, train-test separation, standardization, cross-validation, and a final test set. A stricter design would move interpolation inside each validation fold and would use a chronological test. The largest associations involve $C6H6(GT)$, $PT08.S2(NMHC)$, and $PT08.S1(CO)$. Weather variables have much smaller marginal linear correlations, but nonlinear, delayed, or conditional effects may remain. Among the tested models, KNN with $k = 5$, Manhattan distance, and distance weights gave the best test scores: $RMSE \simeq 0.362$, $MAE \simeq 0.228$, and $R^2 \simeq 0.937$. KNN is the strongest model for this random-split protocol because it captures local structure that the linear models miss. Its local fit is also sensitive to noise, temporal dependence, sensor drift, and distribution shifts. Repeated splits, blocked time-aware validation, and external data are needed before generalization. Ridge and Lasso still provide stable, readable coefficients and form useful reference models. Lasso selected no variables because the penalty was weak and the predictors were highly redundant. The model choice follows the goal: KNN for prediction on similar data, and Ridge for a clearer view of the global linear statistical relations.

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